

Scene-Grounded Inputs

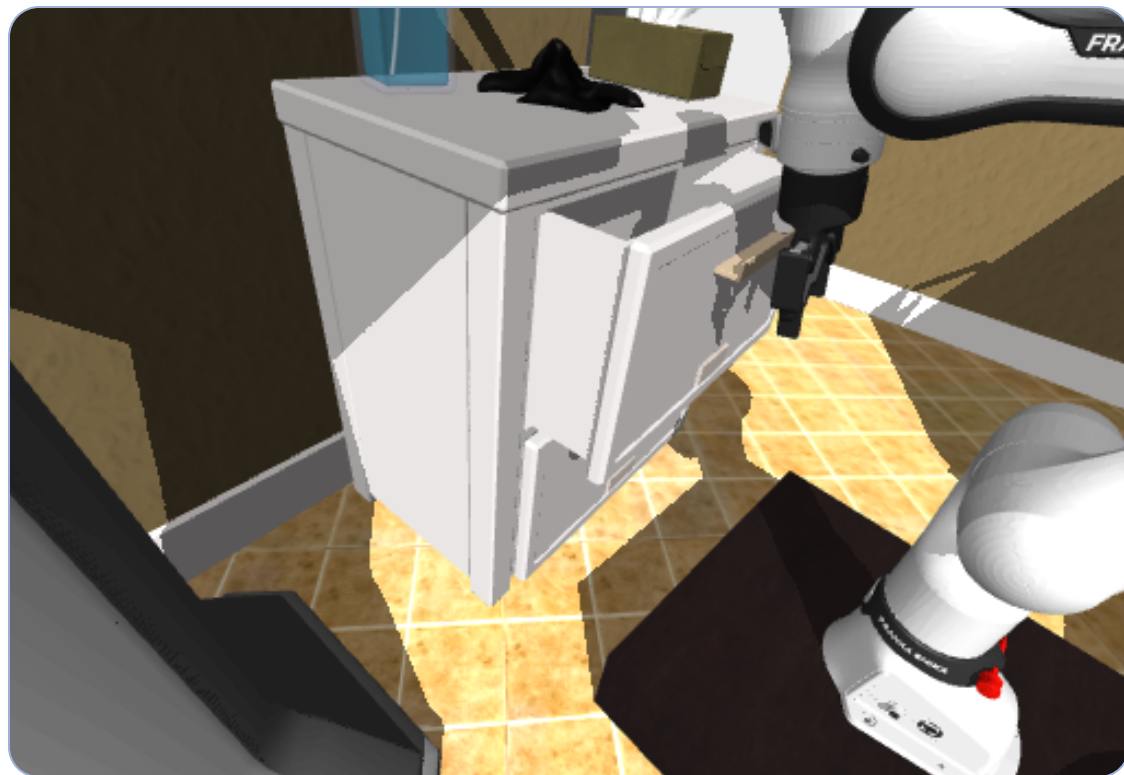
black cloth

brown tissue box

blue spray bottle

white cabinet

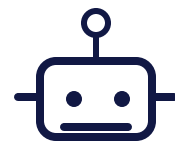
Visible but filtered: [toilet](#) (out of reach)



Recent tasks: 10



Current library:
22 skills (13 primitives + 9 learned)



Task Proposer

Generate $K = 5$ diverse,
scene-grounded objectives

score = novelty $\times$
competence frontier

retry bonus = 0;
failure penalty = 0 for
valid candidates

scene
grounding +
proposer
context



Candidate Ranking ($K = 5$)

1

Close top drawer

0.3600

2

Pull drawer farther open

0.3600

3

Pick black cloth

unsupported family

vetoed

4

Lift brown tissue box

"I'm curious if I can lift the brown
tissue box straight up into the air."

0.6236

SELECTED

5

Place black cloth in drawer

0.1900



Practice selected frontier objective

The attempt exposes a
grasp-centering failure and
motivates a new helper.